ST. MARY'S DSG KLOOF

PHYSICAL SCIENCES PAPER 1 (Physics)

GRADE 10 **TIME: 2 HOURS + 10 min reading**
November 2024 **MARKS: 135**

Examiner: K. Bolton **Moderators: G. Main and T. Kirsten**
NAME: _____

Physical Sciences teacher: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of **20 pages**. Please check that your question paper is complete and write your name on it. A **Data sheet** is also provided.

2. Read the questions carefully.

3. It is in your own interest to write legibly and to present your work neatly.

4. Answer ALL questions on the paper.

5. **All working must be shown** and units may be used in substitutions, but **MUST** be given with answers. Work to 2 decimal places or significant figures where appropriate. Make sure your calculator is in the correct mode.

6. Some extra space on page 20 is included at the end of the question paper. If you run out of space for a question, use this extra space. Clearly indicate the question number of your answer, should you use this extra space.

Question	1	2	3	4	5	6	7	8	Total
Your mark									
Mark	12	15	9	8	15	38	15	23	135

QUESTION 1 – MULTIPLE CHOICE

Answer the multiple-choice questions on the answer grid below. Make a clear cross (X) in the box corresponding to the letter that you consider to be correct. Every question has only one correct answer.

A	B	C	D

Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D

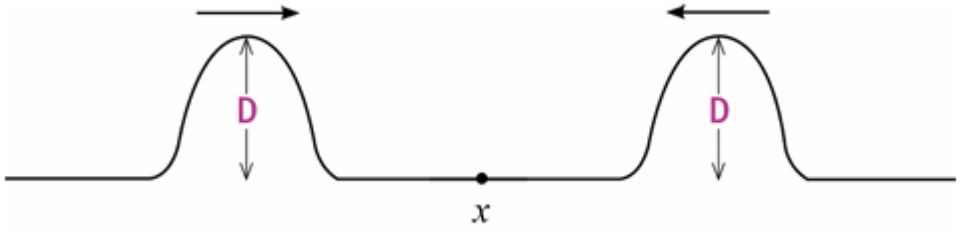
1.1 Sound is:

i a series of moving compressions and rarefactions.
ii an example of a transverse wave.
iii able to travel through a vacuum.
iv an example of a longitudinal wave.

Which of the above statements about sound are correct?

A i, ii and iii
B i and ii only
C i and iii only
D i and iv only

1.2 The sketch shows a rope with two pulses of equal amplitude approaching each other. When the two pulses pass through point X, what is the maximum amplitude of the pulse?



A D
B 0
C 2 D
D ½ D

1.3 Which ONE of the following groups of physical quantities are vectors?

A Acceleration, speed and velocity

B Acceleration, force and velocity

C Mass, distance and acceleration

D Mass, speed and velocity

1.4 An athlete completed one lap of a 400 m oval track in 60 s. The magnitude of her average velocity for the race was:

A 6,67 $m.s^{-1}$.
B 15 $m.s^{-1}$.
C 0 $m.s^{-1}$.
D 4,8 $m.s^{-1}$.

1.5 A car starts from rest and travels in a straight line for 10 s with a uniform acceleration of 2 $m.s^{-2}$. From the end of one second to the end of the next second, the car goes:

A 2 m further than in the previous second.
B 3 m further than in the previous second.
C 6 m further than in the previous second.
D 9 m further than in the previous second.

1.6 A student wants to decrease the resistance of an electrical circuit and has the following options:

(i) Increase the length of the nichrome conductor

(ii) Increase the thickness of the nichrome conductor

(iii) Use a copper conductor instead of a nichrome conductor

Which combination of options would best reduce the resistance of the circuit?

A (i), (ii) and (iii)

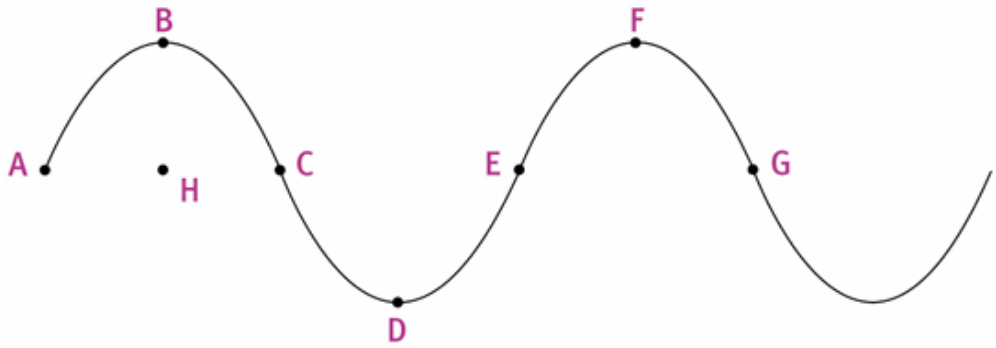
B (i) and (ii)

C (i) and (iii)

D (ii) and (iii)

[6x2=12]

QUESTION 2



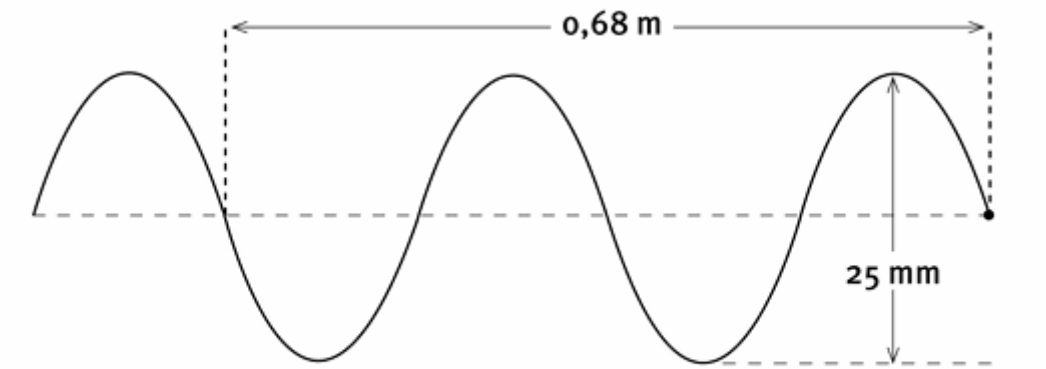
Use the letters supplied in the sketch to indicate the following:

2.1 rest position/equilibrium (1)

2.2 two points that are in phase (2)

2.3 a crest (1)

2.4 Each particle in the wave shown below completes one vibration (oscillation) in 0,4 s.



2.4.1 How long will it take eight wavelengths to pass a specific point in the medium? (2)

2.4.2 What is the amplitude (in cm) of this wave? (2)

2.4.3 What is the period of this wave? (1)

2.4.4 What is the wavelength of this wave? (2)

2.4.5 Calculate the frequency of the wave. (2)

2.4.6 Calculate the speed of the wave. (2)

[15]

QUESTION 3

The diagram below shows the electromagnetic spectrum.



3.1 Give TWO properties that all electromagnetic waves have in common. (2)

3.2 NAME the types of radiation represented by A and B? (2)

3.3 NAME which type of radiation can cause tanning of the skin? (1)

3.4 Which type of radiation next to A in the spectrum has a longer wavelength than A? MICROWAVES or VISIBLE LIGHT. (1)

3.5 Calculate the energy of a microwave quantum of wavelength 0,2 m. (3)

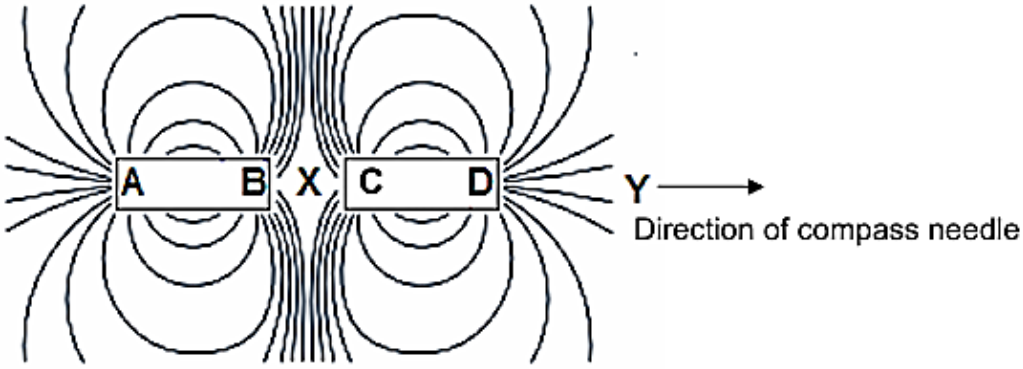
[9]

QUESTION 4

4.1 Define the term *magnetic field.* (2)

4.2 What general term is used to describe materials that are easily magnetised? (1)

The diagram below shows the magnetic field pattern of two identical bar magnets with ends AB and CD.



4.3 What important information about the polarities of B and C is obtained from the diagram? (1)

4.4 A small compass is placed at point Y. The north pole of the compass points AWAY from D.

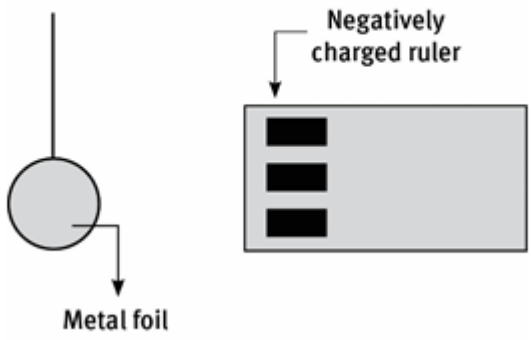
Do the field lines shown around bar magnet CD point from C to D, or from D to C? Give a reason for the answer. (2)

4.5 Give a reason why the Earth's magnetic field is important to our existence. (2)

[8]

QUESTION 5

5.1 A plastic ruler is rubbed vigorously with a cloth. As a result, the ruler has become negatively charged. The ruler is then held close to a small bead covered with thin metal foil, as shown in the sketch.

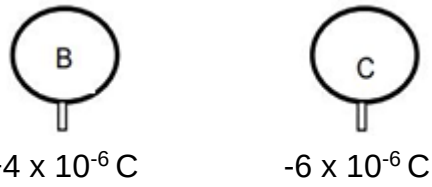


5.1.1 Why has the ruler become negatively charged? (2)

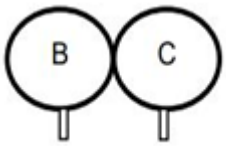
5.1.2 The bead is attracted by the ruler. Explain why it is attracted. (3)

5.1.3 What would you expect to see after the bead has touched the ruler? Explain why this should occur. (2)

5.2 Two identical metal spheres B and C placed on an insulated stand, carry charges of $+4 \times 10^{-6}$ C and -6×10^{-6} C respectively as shown in the diagram below.



The spheres are allowed to touch each other.



After touching, the spheres are then separated and brought back to their original positions as shown in the diagram below.



5.2.1 State the principle of conservation of charge. (2)

5.2.2 Do electrons move from B to C or C to B? (1)

5.2.3 Calculate the new charge on Sphere B and C. (2)

5.2.4 Calculate the number of electrons transferred between the two spheres during contact. (3)

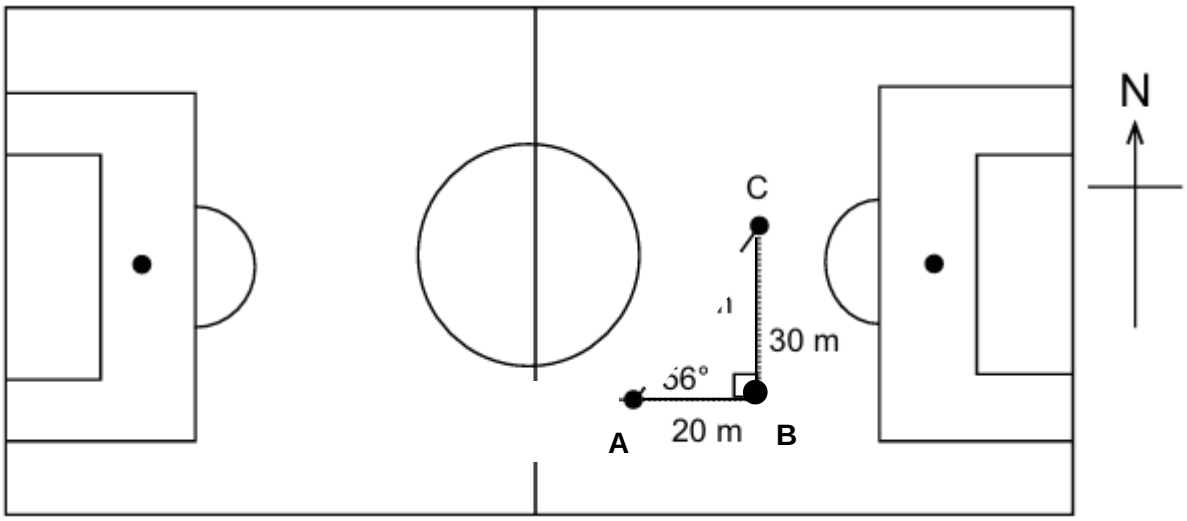
[15]

QUESTION 6

6.1 In a soccer match, a player runs 20 m due East in 10 s from position A to position B (as shown in the diagram below).

He then breaks through the opposition's defence and is able to run to position C in another 10 s.

Position C is 30 m North of B.



6.1.1 Define *resultant vector*. (2)

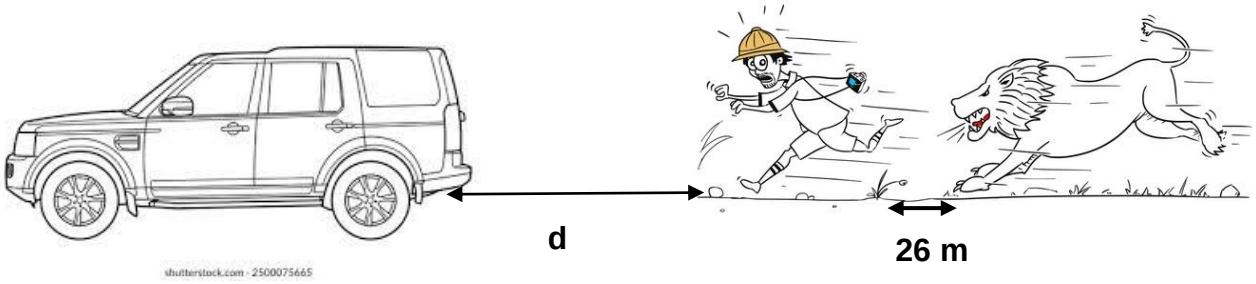
6.1.2 Define *displacement.* (2)

6.1.3 Draw a labelled sketch diagram and determine the player's resultant displacement from position A. (3)

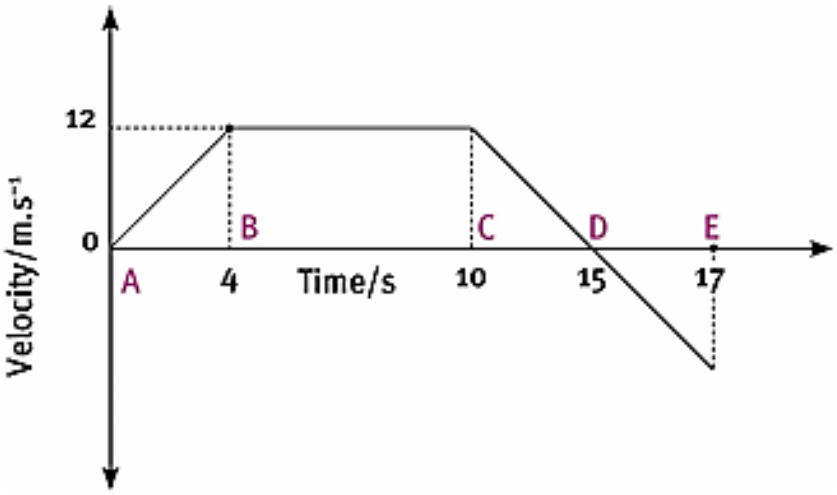
6.1.4 Calculate the player's average speed from A to C. (2)

6.1.5 Calculate the magnitude of the player's average velocity from A to C. (2)

6.2 A person being chased by a lion is running along a straight line towards their car at a constant speed of 4 $m.s^{-1}$. The car is a distance d metres away from the person. The lion is 26 m behind the person and running at a constant speed of 6 $m.s^{-1}$. The person reaches the car safely. What is the maximum possible value of d? (Drawing is NOT to scale). (5)



6.3 The following velocity–time graph represents the motion of a trolley on a track initially moving RIGHT. In carrying out the required calculations DO NOT USE EQUATIONS OF MOTION.



6.3.1 Define *velocity*. (2)

6.3.2 Describe the motion of the trolley for the interval ABC. (4)

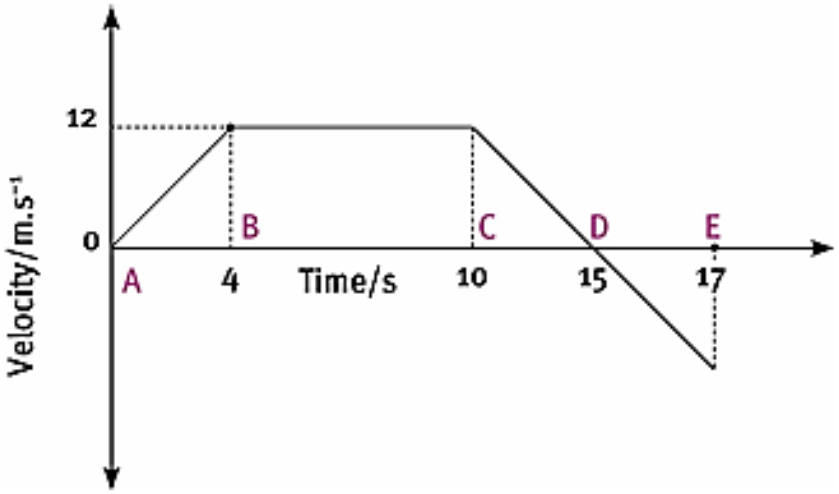
6.3.3 Calculate from the graph the acceleration of the trolley for the time interval CD. (3)

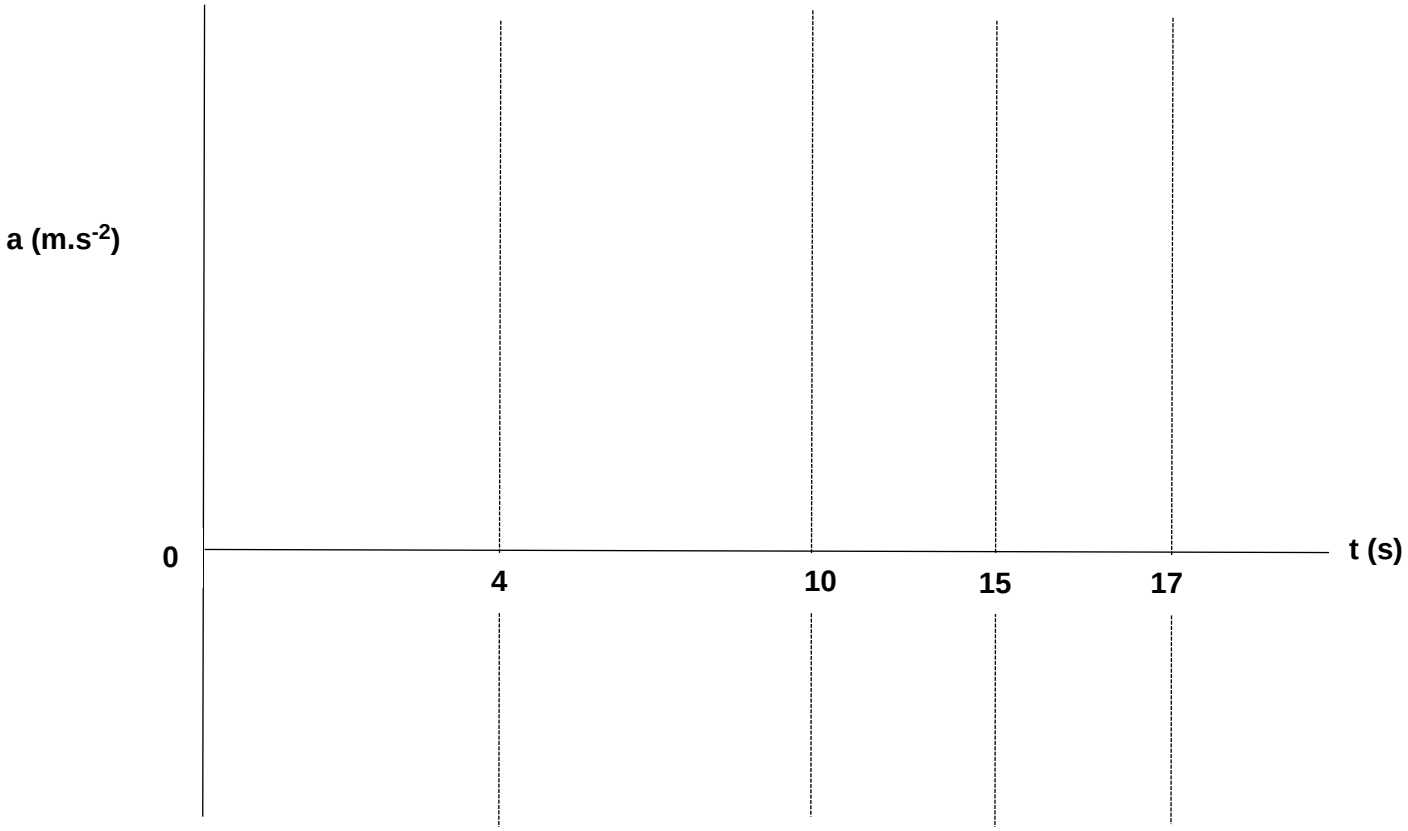
6.3.4 Describe the motion of the trolley for the interval DE? (3)

6.3.5 Calculate the velocity of the trolley at 17 s. (3)

6.3.6 Calculate the displacement of the trolley during the interval A-E. (4)

6.3.7 Draw a sketch of the corresponding acceleration–time graph. Show the time values (0 s to 17 s) on the time axis, but do not show any acceleration values on the Y-axis. (3)

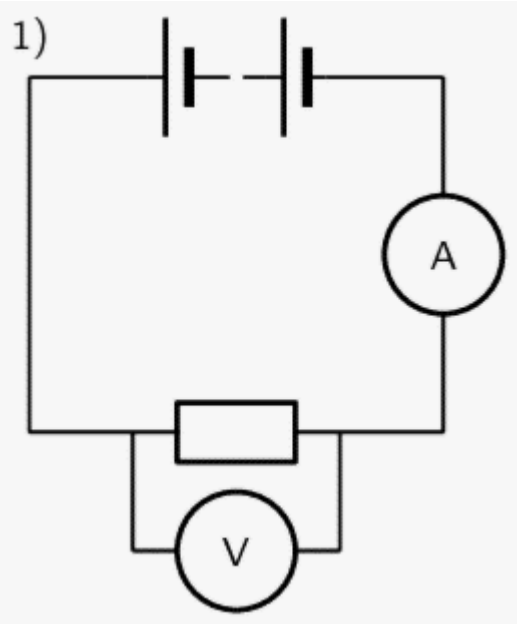




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QUESTION 7

The grade 10 learners want to investigate the relationship between current and potential difference. They vary the applied voltage across the resistor and measure the resulting current through the circuit.



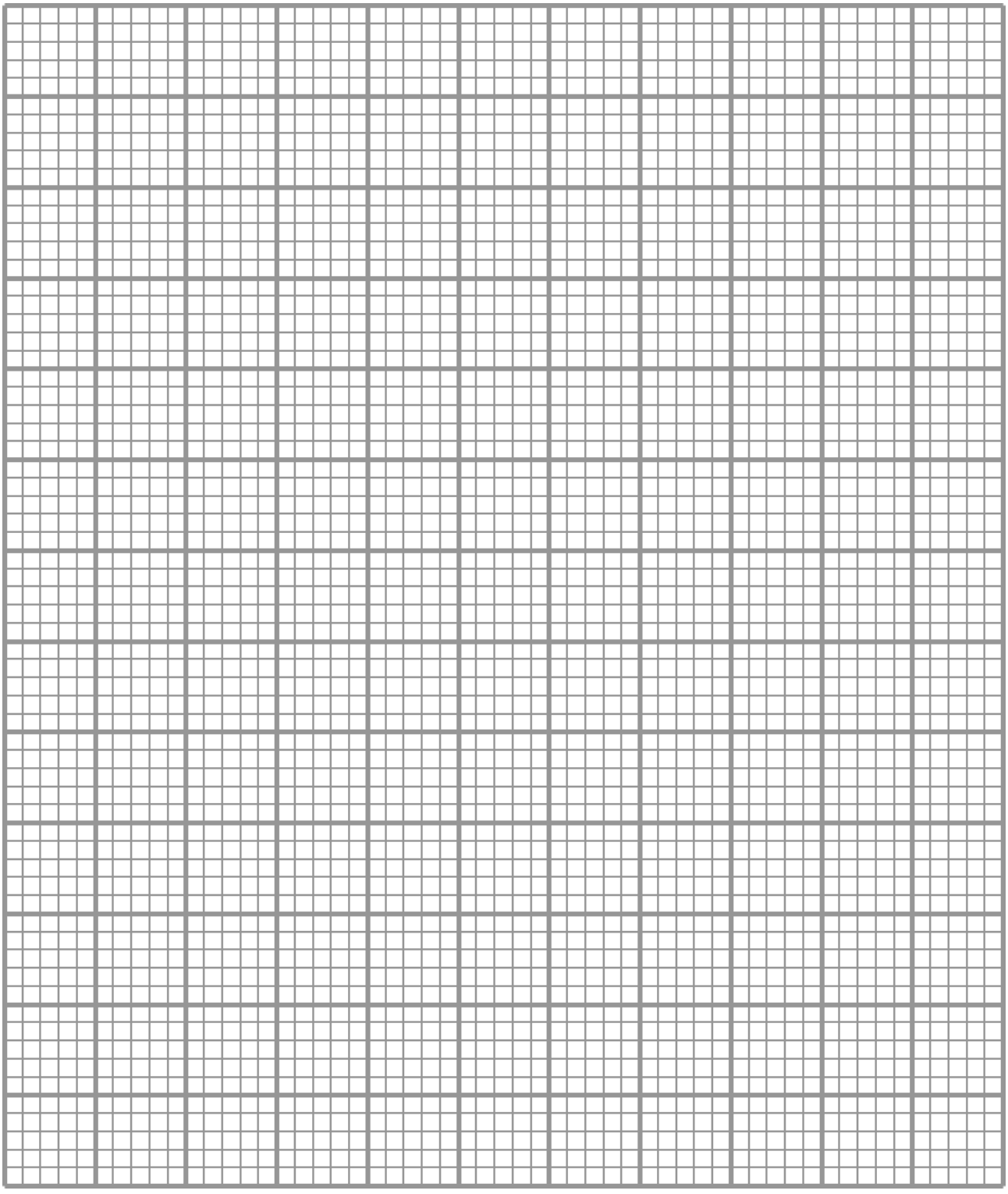
The table below shows the results obtained during the investigation

Voltmeter (V)	Ammeter (A)
1,5	0,7
3,0	1,4
4,5	2,0
6,0	2,5
7,5	3,4

7.1 Name the dependent variable for this investigation. (1)

7.2 What is the control variable? (1)

7.3 Draw a graph (on the paper provided on the next page) of the potential difference versus current. Plot the points and draw a best fit line. (7)



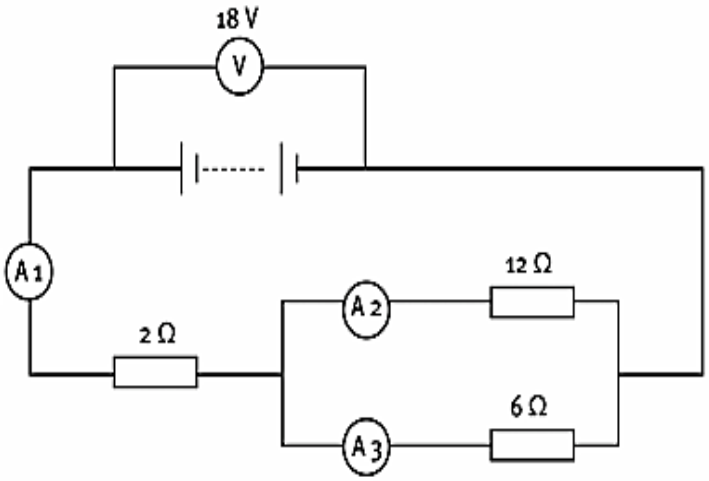
7.4 Calculate the gradient of the graph and hence the resistance of the resistor. (4)

7.5 Using the graph, deduce a relationship that exists between the current and the potential difference in a circuit. (2)

[15]

QUESTION 8

8.1 Study the circuit diagram below and answer the questions that follow.



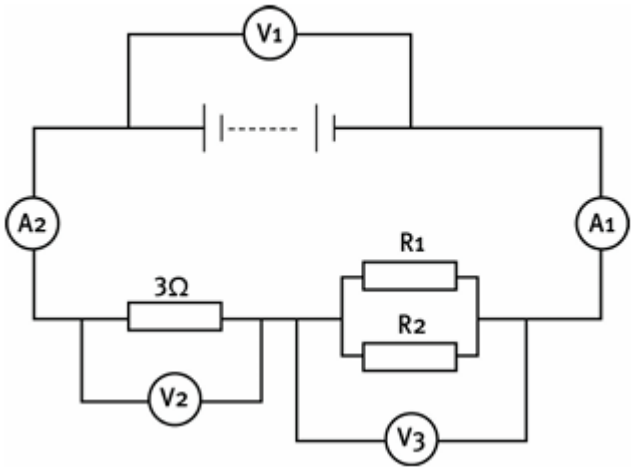
8.1.1 Calculate the equivalent resistance of the whole circuit. (3)

8.1.2 Calculate the reading on A_1. (3)

8.1.3 Calculate the potential difference across the 2 Ω resistor. (2)

8.1.4 Calculate the readings on A_2 and A_3? (4)

8.2 Consider the circuit diagram given below. You are given that V_1 reads 21 V and A_1 reads 3 A. Resistors R_1 and R_2 are IDENTICAL.



8.2.1 Define *potential difference*. (2)

8.2.2 What is the reading on A_2? (1)

8.2.3 Calculate the charge that passes through the 3 Ω resistor in 2 minutes. (3)

8.2.4 Calculate the value of R_1 and R_2. (5)

[23]

EXTRA SPACE: